

Stereotypes, although they have a bad reputation, are in fact crucial to our survival. They tend to simplify things greatly, but they can nonetheless be extremely helpful. In the terms we have been using, a stereotype is a category that, thanks to easily perceived surface-level features, gives us access to a “shallow” kind of depth that has a decent chance of being correct, although the frequency of exceptions is large enough to warrant further refinement of the category. The surface, or more precisely how the surface is perceived, evolves as expertise evolves; as a result, properties that once did not seem to lie at the surface become easily visible, making new kinds of depths available. Experts have categories that evolve over time, allowing them to make observations that are doubly opaque to novices. Firstly, experts are able to see features that elude novices, because what is salient to experts is not salient to novices; secondly, experts associate hidden traits with these subtle surface-level traits, whereas novices almost certainly are totally unaware of those hidden traits.

The reason analogy is so extremely efficient is that appearances are indeed great indicators of essences. This is why reliance on surfaces is not a poor strategy in life. It’s just that in selecting *which* of a situation’s innumerable surface-level features to rely on as clues, one has to do one’s best at separating the wheat from the chaff. This is what the development of expertise does for us. Experts see things that are hidden to novices. They perceive cues that novices either do not see at all or else take for irrelevant, and these surface-level cues give them access to deep perceptions. And thus surfaces become more and more imbued with depth.

Miniature Me-too Stories in a Miniature Domain

There can be no doubt that one of the key questions about human thinking is how we encode situations that we encounter so that later in life they can be spontaneously retrieved from memory. We have just described the life-and-death consequences of how episodes in the global realm of world politics get encoded (and later retrieved) in the minds of decision-makers, and in Chapter 3 we spoke of the same issue but in humbler contexts, such as when a father is watching his one-year-old son engrossed in play with ants and leaves when at the side of the Grand Canyon. We are now going to take another look at the same vital issue, but in a much humbler context — an artificial domain that, some three decades ago, was carved out in order to lay bare many of the central issues of cognition in the clearest possible way.

In Chapter 3, we spoke of the *me-too* phenomenon, typified by cases where you tell a story and then a friend spontaneously reacts, “Exactly the same thing happened to me!” Ironically, these words are a clear tip-off that it was a quite *different* thing that happened to your friend, since what you hear your friend tell is a story involving a different place, a different time, different people, different events, and different words — and yet despite all these differences, you know perfectly well why your friend said, “Exactly the same thing happened to me!” Although on one level their story was totally unlike yours, on another level, a more abstract one, the two are indeed the same. One and the same conceptual skeleton can describe two very different events.

We will now look once again at the me-too phenomenon — this time in the austere microdomain of Copycat. The domain’s innocent-sounding name comes from the fact that when children play at being copycats, their imitations of each other often wind up being surprisingly flexible and creative. It might seem on first glance that being a copycat is nearly mechanical, requiring no imagination or fantasy, but that turns out to be far from the case. For example, if five-year-old Cora waggles her ponytail with her hand, what should six-year-old Xavier, who has no ponytail, do? Well, he could pull a lock of hair just above his forehead, or he might even tweak his nose. If Cora fiddles with one of the buttons on her blouse, Xavier might slide the zipper on the front of his sweater up and down. If Cora removes a barrette from her hair, maybe Xavier, who has no barrettes in his hair, will take off his pair of glasses — and so on. In short, there is plenty of room for verve and playfulness in the act of being a “mere” copycat.

As for the Copycat domain, it is focused on short sequences of letters of the alphabet, dubbed “strings”, as well as on tiny “events” that can happen to these strings — that is, changes that might take place to denizens of that mini-world. Let’s plunge right in with a sample me-too in the Copycat world. To make it come alive a little bit, we’ll pretend that the letter strings can talk. We tune in just as the string *abc* is telling some friends (various letter strings) about the time it got changed to *abd*. Among the listeners is *pqrs*, who pipes up, “Hmm, that’s funny — exactly the same thing happened to me the other day.” Then when *pqrs* tells its story, it turns out that it got changed to *pqrt*. And so... was it *exactly the same thing*, or was it a quite different thing?

If someone wished to see the two micro-events as different, all they would need to do is point out that *abc* contains three letters while *pqrs* contains four, that the two strings have no letters in common, and lastly that putting a *d* in the third position has nothing to do with putting a *t* in the fourth position. Therefore, these are totally dissimilar events! Surely that would burst the balloon of the “exactly the same” claim.

And yet it’s just as easy to flip perspective, to ignore nearly all details, and to declare, “What happened to *abc* was that its rightmost letter got replaced by its alphabetical successor, and that’s also what happened to *pqrs*.” From that perspective, thanks to a little abstraction, the two strings, though of different lengths and having no letters in common, underwent exactly the same change, and so, yes, their two stories were “exactly the same”.

What Gets Encoded When an Event Takes Place?

As we have just seen, “same” or “different” is in the eye of the beholder, and it all depends on what you attach importance to, how much importance you attach to it, and what you consider to be irrelevant. In the real world, we can’t possibly take everything into account all the way down to its most microscopic details, and so we necessarily must ignore almost everything about every situation that we encounter, and that means we unconsciously make a highly selective encoding of it when we store it in memory. We have to strip everything that we experience down to a caricature of itself. The same idea holds in the Copycat microworld.

To be sure, the changing of *pqrs* into *pqrt* is less complicated than the plot of *War and Peace* or than an embarrassing case of mistaken identity at the grocery store, but even in the microworld, the drastic-simplification principle just uttered applies. If you were *pqrs*, you would probably have encoded the event that happened to you in a more abstract fashion than simply storing in your memory the “raw film”, which would merely record the concrete facts in their most boring details — “My entirety was replaced by *pqrt*” — without making any effort to see the *essence* of what took place, which is that most of *pqrs* was left completely alone and only a small part of it changed, and moreover that that part wasn’t a random part but an *extremity*, and it didn’t change in an arbitrary way but in a somewhat *natural* way, which is to say, into a closely related object in a canonical and universally memorized sequence (namely, the alphabet).

In short, you would most likely have (unconsciously) encoded this event in your memory in something like the way we stated above: “My rightmost letter got replaced by its successor” (note that this ignores the identity of the letters in the string; that level of concreteness is seen as irrelevant). Later, when you heard *abc* telling its exciting story of being changed into *abd*, you would encode *abc*’s story in the same way, extracting from it the same conceptual skeleton; this explains why your own story from the past (*pqrs* $\Rightarrow$ *pqrt*) would come leaping out at you when you heard your friend’s more recent tale. For you, hearing *abc*’s story of turning into *abd* would nearly be a case of *déjà vu*.

Of course this little episode is just the tip of the iceberg of the me-too phenomenon in the Copycat world. In order to show how the Copycat domain explores and sheds light on the me-too phenomenon (and thus on the mystery of encoding) in deeper ways, we will now examine a number of more complex examples.

How Humans Do Not Perceive Situations

What event would instantly spring to the mind of *thy* when it hears *abc* tell of being changed into *abd*? Surely the time it was changed into *tkz*, no? That seems like a trivial Copycat challenge, almost the same as the question “What should *pqrs* do, in playing the copycat game, if *abc* turns into *abd*?” And yet *thy* is not a nice tight segment of the alphabet like *abc* or *pqrs*, and therefore if *being an alphabetical segment* is part of the way we see and remember *abc*’s story, then *thy* doesn’t align at all with the stored memory, and so we would have to punt on the Copycat challenge, answering, “The conceptual skeleton that covered *abc*’s change to *abd* is unable to stretch to cover *thy*, because *thy* is not a string of alphabetic successors.” People don’t react that way, and that’s a good thing: we would be very uninsightful beings if our brains came up with conceptual skeletons that were so narrow and so rigid.

Let’s suppose, now, that *ijjkk* was also listening when *abc* told its story of getting changed to *abd*, and that *ijjkk*, too, responded by saying, “Gee, the same thing happened to me one time!” What do you think happened to *ijjkk*? It’s most unlikely that you think that it got turned into *abd*. But why is it so improbable that such a thought would ever have crossed your mind? Because we humans do not like looking at events in the world in such a literal-minded fashion; we prefer *fluid* analogies by far.

In our many years of asking people such questions, no one has ever replied, “Oh, that’s easy — just change *ijjkk* to *abd*!” People spontaneously move away from the literal level; they systematically seek a higher abstraction. Thus in this case, having seen that *abc* changed only at its right end, virtually all people instantly look just at the right end of *ijjkk*, hoping to “do the same thing” in that spot. It doesn’t occur to people to replace the *entire* string *ijjkk* by something else; instead, they seek to replace just a small part of it, since just a small part of *abc* was affected by the original change. In short, people don’t see *abc* as having been replaced lock, stock, and barrel by *abd*; rather, they feel that just the *c* was affected. And they don’t think of that letter as “the *c*”, either, but as “the rightmost letter”, even though they may well *refer* to it as “the *c*”. What they really mean is not the letter itself, but *the role that it plays*.

So this gives us some clues as to how to do “the same thing” to *ijjkk*. We look at its rightmost letter (that is to say, its *c*), which is *k*, and we replace the *k* by its alphabetical successor, namely *l*. This will give us *ijjkl*. Exactly the same thing!

Do you disagree? We certainly hope so. Or to go back to an earlier way of talking about this, we certainly hope that when *ijjkk* is given the chance to spell out how “exactly the same thing happened to me one time”, it doesn’t say, “I got changed into *ijjkl*.” To be sure, that would in *some* sense be “exactly the same thing”, but it would be an impoverished sense of “the same”. We would feel much happier if *ijjkk* said, “The exact same thing happened to me the other day — I got changed into *ijjll*.” Now why does this version seem so much more satisfying to us?

The Inescapable Role of Esthetics

In a sense, the event $ijjkk \Rightarrow ijjkl$ is “exactly the same” as the event $abc \Rightarrow abd$, yet in a deeper sense, the event $ijjkk \Rightarrow ijjll$ is even *more* exactly the same as $abc \Rightarrow abd$. This is an esthetic judgment, and when simple, basic perceptions are concerned, there is a widely shared sense of esthetics that depends on many factors, including abstraction and frame-blending (discussed below), and that sense is crucial to how we humans perceive the world around us. There are virtually universal mechanisms that guide us in our perceptions of the world, and that’s lucky, because it means that in many ways, two people (or even a whole crowd of people) will be likely to see what happens in a given situation in the same way, even though *in theory* they all could have focused on entirely different aspects of the situation, and could therefore have encoded the situation in wildly different ways. In many simple situations, there is a strong natural tendency to see things in just one way and one of the beauties of the Copycat domain is that it helps us put our finger on some of what these nearly universal tendencies are.

What lies behind the answer *ijjll* is an almost irresistible tendency for people to *make perceptual chunks*. It is thanks to gestalt psychology, developed in the early part of the twentieth century and particularly influential in the 1930s, that today everyone takes for granted the importance of such perceptual chunking. But it was not always so obvious. Even if gestalt psychology has often been criticized for merely describing rather than explaining mental phenomena, the importance of its findings has never

been questioned. Certain perceptual principles, including that of “continuity”, of “good form”, and the motto that “the whole is greater than the sum of its parts” (which has to do with our ability to perceive large-scale patterns), come directly from gestalt psychology. For our purposes, what matters most is the idea that certain ways of making perceptual chunks strike most people as natural — indeed, as well-nigh intrinsic — ways of apprehending the world.

Let’s return to the case of $ijjkk \Rightarrow ijll$. Instead of seeing the original string $ijjkk$ as a string of six letters, most people tend to see it as a string of *three pairs*: $i\text{--}j\text{--}kk$. This interpretation invites us to map these three pairs onto the three letters of abc . Here something subtle happens. Because of the perceptual chunking that took place, the rightmost “letter” of $i\text{--}j\text{--}kk$ is no longer the single letter k but the *group* of two k ’s. In other words, the meaning of the word “letter” is extending naturally outwards under contextual pressure. In this case, we recoil at the idea of merely changing the single rightmost k into an l ; instead we change *both* of the k ’s simultaneously into l ’s, yielding the string $ijll$. In sum, we changed the rightmost “letter” to its successor, but in so doing, we instinctively took the notion of *letter* fluidly and abstractly.

The story behind the answer $ijll$ will not be a surprise to you, but the real point is how intimately and inextricably wound up in all of this our sense of esthetics is, and how often people see eye-to-eye on the esthetic qualities of different answers.

Let us be even more explicit here. In a strict sense, there is nothing *wrong* with the answer $ijjkl$ (it clearly has a certain kind of logic to it), nor for that matter is there anything wrong with $ijjkd$ (“replace the rightmost letter by d ”), nor is there even anything wrong with the answer abd (“replace the entire string by abd ”). We are not concerned with *rightness* or *wrongness*, *validity* or *truth* here. This is not a black-and-white domain in which there is just one right answer and all other possibilities are dead wrong; rather, there is a spectrum of possible answers, and each different answer has its own logic, and answers vary in their degree of appeal. *Appeal*, not truth, is the name of the game. Rather than being *wrong*, the various answers discussed above might be said to have differing “scores” of subtlety or finesse, depending on how much or how little abstraction was involved, as well as on an elusive “sense of essence”. Especially the final pair ($ijjkd$ and abd) are out of tune with how humans tend to see their world.

The Left Hand Doesn’t Know What the Right Hand is Doing

To shed some perspective on all this, we propose a simple analogy challenge. First please hold up your two hands, with their palms facing you, and wiggle your right thumb. Now we’ll simply ask your left hand to “Do the same thing.”

So you wiggled your left thumb? Good move! But you certainly could have wiggled your left hand’s little finger instead, since it is that hand’s *rightmost* finger, and since your right thumb is your right hand’s rightmost finger. However, what you *could have done* doesn’t necessarily have much to do with what people in fact *do*, because we humans are ceaselessly categorizing the world, and we try to do so efficiently and even *elegantly*. We don’t usually see a little finger as being analogous to a thumb, unless there